

# A Verification-Gated Engine for Geometry→Arithmetic Correspondences, and the (O2) Negative

(VFD programme — working draft)

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## Abstract

We describe a reproducible, gate-first engine for testing proposed correspondences between geometric objects and arithmetic  $L$ -functions, and we use it to settle one specific question in the negative. A proposed correspondence (“expression”) is accepted only if it passes an eight-check verification gate—chief among them a *provenance* check that rejects any map fitted to its target, and a battery of spectral discriminators (zero positions, density, nearest-neighbour spacing, number variance) that separate one  $L$ -function from another. Applied to the VFD programme, the engine *certifies* that the icosian substrate realizes  $\zeta_K$  (and  $L(\chi_5)$ ,  $\zeta(s)\zeta(s-1)$ ), and *rejects*, on five independent grounds, the stronger claim that it realizes the Riemann  $\zeta$  itself. We call this the (O2) negative. The engine issues verification certificates only and makes no claim about the Riemann Hypothesis.

## 1 Motivation: the hardcoding trap

Claims of the form “geometric object  $X$  realizes arithmetic object  $Y$ ” are easy to make and easy to fake: one may fit a map from  $X$ ’s data to  $Y$ ’s known invariants and then “recover”  $Y$ . Such a map encodes nothing—it is circular. A prior internal audit of this programme (the “W5” audit) caught exactly such a fitted substrate→eigenvalue map. The purpose of the present engine is to make the verification, rather than the proposal, the primary and reproducible step: a correspondence is a *result* only if it survives an explicit gate that a fitted or overclaimed map cannot.

## 2 The engine

An *expression* is a tuple (geometry, claim, predict, meta, provenance), where **predict** is a parameter-free procedure that produces the on-line zeros of the claimed  $L$ -function *without taking the target as input*. The gate evaluates eight checks:

1. **no-hardcoding**: the target is not an input to **predict**;
2. **provenance**: no constant is fitted or tuned to the target;
3. **fingerprint**: nearest-neighbour spacing (fraction of gaps  $< 0.3$ ), which separates a single  $L$ -function ( $\approx 0$ , GUE repulsion) from a superposition;
4. **pointwise**: agreement of the first  $k$  zeros;
5. **degree** and **pole**: structural metadata match;
6. **density**: zeros-per-unit-height (Weyl rate);

7. **rigidity**: number variance  $\Sigma^2(L)$ .

An expression is **verified** iff it passes all applicable checks; a certificate (a hashed record of every check and its evidence) is emitted either way. A proposer interface (“ARIA”) may suggest expressions; it never certifies—the gate does. The engine is deliberately a *certifier*, not a search: most honest outputs are negatives.

### 3 The certified atlas

Running the engine across the programme’s correspondences yields:

**Certified Result 1.** The following are **verified** (all eight checks): the icosian/ $V_{600}$  substrate  $\leftrightarrow \zeta_K(s)\zeta_K(s-1)$ ; the 24-cell  $\leftrightarrow \zeta(s)\zeta(s-1)$ ; the  $\varphi$ -shell  $\leftrightarrow L(\chi_5)$ .

**Certified Result 2** (the  $(O2)$  negative). The proposed correspondence “substrate  $\leftrightarrow \zeta$ ” is **rejected** on five independent checks: fingerprint (0.077 vs. 0.0), pointwise (first-zero displacement  $\approx 22$ ), density, rigidity, and degree (4 vs. 1). The substrate realizes  $\zeta_K = \zeta \cdot L(\chi_5)$ , which *contains*  $\zeta$  as a factor but does not *isolate* it.

**Certified Result 3** ( $(O2)$ , independent second route). The negative is confirmed by a route disjoint from the spacing discriminators: a direct *count* of on-line zeros. Since  $\zeta_K = \zeta \cdot L(\chi_5)$ , the substrate’s  $L$ -function carries, on the critical line, the union of the zeros of  $\zeta$  and of  $L(\chi_5)$ ; its on-line zero density therefore strictly exceeds that of  $\zeta$  (empirically a factor  $\approx 3$ –4.7 up to height 50, tending to 2 asymptotically as each factor contributes  $\sim \frac{1}{2\pi} \log \frac{T}{2\pi}$ ). A spectrum strictly denser than  $\zeta$ ’s is not  $\zeta$ ’s.

The two routes—spacing/fingerprint (Result 2) and counting/density (Result 3)—are independent and agree.

Two controls confirm the gate’s discrimination:

**Certified Result 4** (fitted-map control). A map declared as least-squares fitted to the zeros is rejected on *provenance* alone, regardless of how well its output matches—reproducing the  $W5$  catch mechanically.

**Certified Result 5** (class-vs-individual control). A generic GUE operator (random Hermitian, no arithmetic) *passes* fingerprint and rigidity (it has the right universality class) but *fails* pointwise and density (it is not the right individual). Thus GUE-class membership is certified to be insufficient to be  $\zeta$ .

| expression                               | verdict  | failing checks (cert id)                                        |
|------------------------------------------|----------|-----------------------------------------------------------------|
| 24-cell $\rightarrow \zeta(s)\zeta(s-1)$ | VERIFIED | — (2c9f2a5a)                                                    |
| icosian $\rightarrow \zeta_K$            | VERIFIED | — (0ae81ff8)                                                    |
| $\varphi$ -shell $\rightarrow L(\chi_5)$ | VERIFIED | — (080d0d3c)                                                    |
| substrate $\rightarrow \zeta$ (bridge)   | REJECTED | fingerprint, pointwise, density,<br>rigidity, degree (f5c303f3) |
| fitted map $\rightarrow \zeta$           | REJECTED | provenance (50a16461)                                           |
| generic GUE $\rightarrow \zeta$          | REJECTED | pointwise, density, rigidity<br>(051fff64)                      |

The certified atlas (hashes abbreviated). Each REJECTED entry fails for a reason intrinsic to the discriminator named, not by adjudication.

## 4 Reproducibility and scope

All certificates are content-hashed and emitted to a machine-readable atlas; the engine, gate, and scripts accompany this note. We claim only what the certificates state: which correspondences pass the gate and which do not. In particular, Result 2 is a statement that a specific *construction* fails to realize  $\zeta$ ; it is **not** a statement about the zeros of  $\zeta$ , and implies nothing about the Riemann Hypothesis. No proof certificate is defined or issued.